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February 28, 1989
SUPERSEDING
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May 18, 1984

MILITARY SPECIFICATION

SEMITRAILERS, REFUSE COLLECTION, COMPACTION TYPE: COMMERCIAL

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers commercial, compaction type, refuse collection semitrailers.

1.2 Classification. Semitrailers are of the following types, as specified (see 6.2):

1.2.1 Type. The type of semitrailers will consist of the following:

Type I - See 6.6.
Type II - Tandem axle, 50 cubic yard (cu yd).
Type III - See 6.6.
Type IV - Tandem axle, 65 cu yd.
Type V - Tandem axle, 75 cu yd.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 2330

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Standards. The following standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

FEDERAL

ZZ-T-381 - Tire, Pneumatic, Vehicular (Highway).

STANDARDS

FEDERAL

FED-STD-297 - Rustproofing of Commercial (nontactical) Vehicles.

MILITARY

MIL-STD-281 - Automobiles, Trucks, Truck Tractors, Trailers and Trailer Dollies; Preservation and Packaging of.

MIL-STD-1223 - Nontactical Wheeled Vehicles; Treatment, Painting, Identification Marking, and Data Plate Standards.

MIL-STD-1474 - Noise Limits for Army Material.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the date of the solicitation.

DEPARTMENT OF TRANSPORTATION (DoT)

Federal Motor Vehicle Safety Standards and Regulations.
Federal Motor Carrier Safety Regulations.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.2 Non-Government publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue

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of the DODISS cited in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B 117 - Salt Spray (Fog) Testing, Standard Method.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE J318 - Air Brake Gladhand Service (Control) and Emergency (Supply) Line Couplers - Trucks, Truck-Tractors, and Trailers, Recommended Practice. (DoD adopted)
SAE J534 - Lubrication Fittings. (DoD adopted)
SAE J560 - Seven-Conductor Electrical Connector for Truck-Trailer Jumper Cable. (DoD adopted)
SAE J588 - Turn Signal Lamps for Use on Motor Vehicles Less than 2032 mm in Overall Width.
SAE J682 - Rear Wheel Splash and Stone Throw Protection, Recommended Practice. (DoD adopted)
SAE J700 - Upper Coupler Kingpin - Commercial Trailers and Semitrailers.
SAE J702 - Brake and Electrical Connection Locations - Truck-Tractor and Truck-Trailer, Recommended Practice. (DoD adopted)
SAE J1292 - Automobile, Truck, Truck-Tractor, Trailer, and Motor Coach Wiring, Recommended Practice. (DoD adopted)

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

TIRE AND RIM ASSOCIATION, INC. (TRA)

TRA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The semitrailer shall be a compaction type equipped with an all metal box type construction body, forward hopper area, gasoline or diesel-engine-driven hydraulic pump system and rear discharge.

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3.1.1 Standard commercial product. The semitrailer shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the semitrailer being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.2 First production vehicle. The contractor shall furnish a semitrailer for first production vehicle inspection.

3.3 General design.

3.3.1 Federal motor vehicle safety standards. The semitrailer shall comply with DoT Federal Motor Vehicle Safety Standards and Regulations in effect at time of manufacture.

3.3.2 Net weight. The net weight of the fully equipped semitrailer shall be the minimum practicable for the service intended.

3.3.3 Gross vehicle weight (gvw). The gvw shall consist of the net weight and the payload, as specified in 3.3.4, evenly distributed within the load area.

3.3.4 Volumes and payloads. Volumes and payloads shall conform to table I for specified type of semitrailer.

TABLE I. Volumes and payloads.

	Type		
	II	IV	V
Net volume of refuse body, cu yd (not less than)	50	65	75
Volume of compaction portion cu yd (not less than)	34	49	55
Refuse payload capacity, lb (not less than)	24,000	36,000	36,000

3.3.5 Weight distribution. The proportion of the gvw to be supported on axle suspensions, measured at the ground, shall not exceed 18,000 pounds (lb) per axle.

3.3.6 Dimensions and clearances. The semitrailer, uncoupled from truck tractor, resting level on landing legs on level ground, and without payload, shall conform to the dimensions and clearances specified in table II.

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3.4 Performance. The semitrailer, fully equipped and loaded with the specified payload (see 3.3.4), shall be capable of being towed at speeds of not less than 10 miles per hour (mph) over unimproved roads (see 6.5), and over reasonably hard uneven terrain; and of being towed at speeds of not less than 55 mph over improved roads. The semitrailer shall be capable of traveling 55 mph, unloaded, over improved roads.

3.4.1 Turning ability. The semitrailer shall be capable of assuming a 90 degree angle to the coupled towing vehicle without cramping, or damage to the semitrailer or the towing vehicle.

TABLE II. Dimensions and clearances.

	Type		
	II	IV	V
Overall length, inches (not more than)	480	480	480
Overall width, inches (not more than)	96	96	96
Overall height, inches (not more than)	150	156	162
Ground clearance, inches (not less than)	9	9	9
Swing radius, from centerline of kingpin to most distant point on semitrailer nose, inches (not more than)	52	60	60
Distance from centerline of kingpin to front end of semitrailer, inches (not less than)	18	18	18
Turning clearance from the centerline of kingpin to any portion of semitrailer 6 inches or more below upper fifth wheel plate, inches (not less than)	78	80	80
Upper fifth wheel height, from the ground to underside of plate, inches ±1 inch	48	48	48

3.4.2 Tracking ability. The semitrailer shall conform to the tracking requirements of DoT Federal Motor Carrier Safety Regulations, section 393.70(a).

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3.4.3 Brake performance. The service brakes shall stop the tractor-semitrailer combination, with the semitrailer loaded with the specified payload, within the stopping distance requirements of Federal Motor Carrier Safety Regulation 393.52.

3.4.4 Accessibility. The design of the vehicle and optional equipment shall permit access for routine servicing and shall permit access for replacement and adjustment of component parts and accessories with minimal disturbance of other components and systems.

3.5 Chassis frame and body structure. Unless otherwise specified (see 6.2), the semitrailer shall be the chassis frame type. Frame and body shall be an electrically welded, all-steel, general constructed, integral unit. The rated payload (see 3.3.4), and the sprung weight of the semitrailer shall not impose a fiber stress greater than 50 percent of the yield strength of the materials used when the semitrailer is operating under mobile operating conditions specified in 3.4. The chassis main frame shall extend the full length of the body and may be used as guidance members for the packer plate, shall be combined with sufficient number of cross members to insure stability, and may be spliced at the front to provide a gooseneck platform for mounting the auxiliary power source.

3.5.1 Body. The body configuration shall be the manufacturer's standard box shape with the hopper (loading space) and opening located at the forward end of the semitrailer. The body sides, top, and forward end shall be fabricated of not less than 0.1046-inch (U.S. revised standard gage No. 12) thick high tensile steel, and the floor fabricated of not less than 0.1345-inch (U.S. revised standard gage No. 10) thick high tensile steel. The floor area beneath the hopper opening shall be provided with support equivalent to 3-inch pressed steel or structural steel channels placed on centers not exceeding 12 inches, and shall withstand a load of not less than 500 pounds per square foot area without deformation. The compaction area, extending from the rear of the hopper opening to the rear of body, shall withstand the maximum pressure exerted by the packer plate upon the refuse without permanent deformation. The edges around the hopper opening shall be framed and reinforced to resist denting by heavy objects dumped inside and withstand bumps encountered during loading operations. The interior surfaces of the body shall be smooth to expedite ejection and cleaning operations. The body shall have a minimum of two rows of retaining teeth in the top inside of the body to prevent rubbish from falling forward after compaction stroke. A means shall be provided for prevention of immobilization of the packer due to wedging of materials between the top of the packer plate and the roof of the body. The entire body shall be liquid-tight to a height of not less than 5 inches.

3.5.1.1 Hopper. The hopper opening shall extend across the top and straight downward along each side to provide a loading height of not more than 120 inches in order to facilitate loading from either side of semitrailer. Side openings shall be the full length of the hopper and may be accomplished by means of hinged doors. The capacity of the hopper shall be such that the contents of a standard refuse container of not less than 12 cu yd volume be emptied without overflow to the outside and without shaking, packing, tamping, or undue movement of either semitrailer, container, or container handling device. Unless otherwise specified (see 6.2), the size of the hopper opening shall be not less than 80 inches wide by 100 inches long.

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3.5.1.2 Hopper cover(s). Unless otherwise specified (see 6.2), power operated hopper cover(s) which completely cover the hopper opening and remain tightly closed during vehicle travel shall be provided. When power operated cover or covers are furnished, the open or close cycle shall be not more than 30 seconds.

3.5.1.3 Rear discharge door(s). The top or hanged rear discharge door(s) shall open an unrestricted opening for ejecting the refuse. The door(s) shall be reinforced to prevent deformation which may cause leaks between door(s) and body during packing operation. The rear door(s) shall be provided with heavy-duty cam type or wedge type latches and heavy-duty hinges. The latches shall hold the door(s) securely in locked position. The latching mechanism shall be actuated by a single lever except that, when double doors are provided, a single lever actuator may be used for each door. Latch actuating levers shall be readily operable by a man standing on the ground. The hinges shall keep the door(s) tightly alined to prevent leakage resulting from packer plate compacting the refuse. When power operated top hinged discharge door is furnished, the open or close cycle shall be not more than 30 seconds.

3.5.1.4 Discharge spillage. When necessary, provisions shall be provided to prevent discharge refuse from spilling forward around wheels and collecting on suspension components.

3.5.1.5 Packer plate. The packer plate shall be actuated by double acting, telescopic, hydraulic cylinder assembly, have a travel length extending from the forward end of the hopper to the rear discharge doors(s). The packer plate shall be so shaped to cause the packing side to force the refuse to be simultaneously compacted rearward and upward, closely fit the cross-sectional dimensions of the compacting area to preclude bypassing refuse, and be guided to prevent scraping and gouging of the body inside walls. When maximum pressure is applied or released, no permanent misalignment shall be evident.

3.5.1.6 Packer plate operation. The packer plate operation for compacting and ejecting refuse shall conform to the following:

a. The complete compaction cycle, comprised of the packer plate movement from the initial station at the loading hopper to any intermediate position and return, shall be accomplished in not more than 2 minutes.

b. The complete ejection cycle, composed of the packer plate movement from the initial station at the loading hopper to the final location at the discharge door opening and return, shall be achieved in not more than 2.5 minutes.

c. The packer plate unit pressure force range shall be from a minimum of 2.5 pound-force per square inch (psi) up to not less than 8.5 psi predicted upon variance in force (product of pressure and area) exerted by the smallest and largest inside diameter (proportionate area) cylinders of the telescoping hydraulic cylinder assembly, operating pressure of the hydraulic pump less head losses, and packer plate area.

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d. The packer plate shall completely expel the total volume (maximum rated payload) of refuse without external assistance (manual raking or pulling of body contents).

e. The packer plate action, from the extreme final position at the loading hopper to the extreme final position at the discharge door(s), shall be achieved in a single continuous sweep.

3.5.2 Power source. Power source for hydraulic equipment operation shall be a standard commercial, water-cooled, gasoline or diesel engine. Engine shall have sufficient horsepower and torque to operate packer plate as specified herein and shall be equipped with all necessary accessories enclosed in a weather-proof housing, including the following: electric starting, high capacity generator or alternator and regulator+, battery, air cleaner, oil filter, muffler, choke, throttle, governor, fuel pump, fuel tank, spark arrester, charging indicator, oil pressure indicator, and coolant temperature indicator. The engine shall be arranged with a device which automatically accelerates the engine speed during packing operations which embodies means for adjusting the degree of acceleration. The noise level generated by the power source shall not exceed 85 decibels (reference pressure 0.0002 microbars) measured at the operator's station as per MIL-STD-1474, category D, or the equipment shall be placarded "Warning - Noise Hazard Area - Hearing Protection required when power unit is in operation." The auxiliary engine exhaust system or the hydraulic system shall be shielded so that, in the event of a hydraulic line rupture or hydraulic fluid leak, the hydraulic fluid will not come into contact with any part of the vehicle exhaust system.

3.5.2.1 Engine hour meter. Unless otherwise specified (see 6.2), an engine hour meter having a totalizing mechanism of not less than 9,999 hours, shall be furnished to register accurately the number of hours of engine operating time. Mete+r shall be of rugged construction to insure continuous trouble-free performance under the most extreme equipment operating conditions indicated in this specification.

3.5.3 Hydraulic system. The hydraulic system shall include at least the components specified herein.

3.5.3.1 Hydraulic pump. The hydraulic pump shall have a rated output capacity (volumetric delivery and operating pressure) to permit fulfilling of the requirements specified in 3.5.1.2, 3.5.1.3, and 3.5.1.6.

3.5.3.2 Protective devices. The hydraulic system shall embrace overload protective device(s) (relief valves) for setting the limit of maximum operating pressure to prevent damaging the constituent hydraulic components. To prevent over travel, limit control devices shall be provided, where necessary to automatically stop the movement of packer plate at both ends of travel.

3.5.3.3 Hydraulic cylinders. The hydraulic cylinders shall be the manufacturer's current standard size (inside diameter and lengths) and fitted with seals to prevent entrance of foreign matter and to preclude leakage of hydraulic fluid. All exposed hydraulic cylinder piston rods, when in the fully extended position, shall be hard chromium plated with sufficient thickness to withstand 30-hour salt spray test in accordance with ASTM B 117.

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3.5.3.3.1 Telescopic cylinders. The double acting, telescopic cylinders for actuating the packer plate shall fulfill the following requirements:

a. Demonstrate continuous sweep action within the allotted time specified in 3.5.1.6(a) (b), and (e).
b. Develop sufficient power to enable the packer plate to exert the pressure force specified in 3.5.1.6(c).
c. Furnished with fixed anchor (adjustable traveling anchor is not acceptable).

3.5.3.3.2 Single cylinders. Each set of double acting single unit cylinder(s) for operating the hopper cover(s) and top hinged discharge door (when furnished), shall meet at least the following requirements:

a. Exhibit continuous movement within the allotted time during open or close cycle specified in 3.5.1.2 and 3.5.1.3.
b. Each cylinder provided with a fixed anchor.

3.5.3.4 Reservoir and filters. The hydraulic fluid reservoir shall have the capacity to maintain the fluid temperature within safe limits to preclude damage to the seals, and the capacity to prevent danger of cavitation. The reservoir shall be equipped with a fluid level gage visible from the operator's station or a reservoir dipstick may be provided. The reservoir shall be free of foreign matter (mill scale and other particles) prior to filling with hydraulic fluid. Means shall be provided for draining, cleaning the interior, and refilling the reservoir. Filters, having capacity to permit unimpeded flow, shall be installed.

3.5.3.4.1 Hydraulic fluid. Hydraulic fluid conforming to the manufacturer's standard or recommended requirements shall be provided in the hydraulic system. A photoetched, corrosion-resistant caution plate, located near the hydraulic system fill port, shall be furnished stating the type of hydraulic fluid required.

3.5.3.5 Pressure gage. When specified (see 6.2), a shock-resistant hydraulic pressure gage shall be permanently installed between the pressure (discharge) side of the hydraulic pump and the pressure (inlet) side of the hydraulic cylinders. The gage shall be visible from the operator's station. Provisions, such as T-fitting with plug, shall be made for installing a pressure gage between the outlet side of the hydraulic cylinders and return line to the fluid reservoir.

3.5.3.6 Hydraulic pressure lines and fittings. All hydraulic pressure lines shall be seamless steel hydraulic tubing or wire braid hoses with permanently attached fittings. All fittings shall conform to the rate of flow and pressures required to operate the hydraulic system smoothly and efficiently.

3.5.3.7 Controls. Either manual lever or electric pushbutton type controls shall be provided to regulate the movements of the hopper cover(s), and top hinged discharge door, when furnished. The packer plate hydraulic cylinder control shall embody pack, neutral, and return positions. All controls shall automatically return to neutral position when released. When manual lever type controls are furnished, the control movements shall coincide with the packer

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plate movement if possible. The manual or electrical controls shall be arranged to permit operations from both stations, one at engine platform level, and one at elevated level which permits operator to view movement of packer plate. All controls shall be identified as to function(s).

3.5.4 Sump tank. When specified (see 6.2), a liquid collection tank, of not less than 16-gallon capacity, shall be provided under the body to catch drainage from wet refuse. A large faucet, for handling heavy liquids, shall be provided to drain the tank. Sump tank need not be provided when the same provision is incorporated in the semitrailer body.

3.5.5 Washout system. When specified (see 6.2), a washout system shall be furnished with a reservoir of not less than 80-gallon capacity. When compressed air is utilized to provide discharge pressure, reservoir shall be a pressure vessel designed for operating pressure of not less than 100 psi. The air system shall be provided with a safety relief valve, dial pressure gage, air charging valve, and bleed valve.

3.5.6 Access ladders, platforms, handholds, and railings. Access ladders, platforms, and handholds shall be installed at the front of the body to provide access to engine, top of body, and hopper. Platforms shall have nonslip tread surfaces. Railings shall be provided around platform.

3.5.7 Suspension system. The semitrailer shall be furnished with the manufacturer's standard suspension system. Each component of the suspension system shall have a rated capacity at least equal to the imposed load when loaded with its specified payload (see 3.3.4). Clearances shall preclude interference between wheels and with any other part of the semitrailer under the operating conditions specified herein.

3.5.8 Axles. Semitrailer shall be furnished with a tandem axle. Axle ratings shall be at least equal to the load imposed on each axle, measured at the ground when the semitrailer is loaded with the applicable specified payload. The wheel bearings and axle spindles shall be oil lubricated. The oil viscosity shall be in accordance with the manufacturer's recommendations. The hub caps shall have a body and a window for visual determination of oil level. Provision for venting, or equivalent method of withstanding pressures without leakage, and for replenishing the oil supply shall be incorporated.

3.5.9 Wheels, rims, tires, and inner tubes.

3.5.9.1 Wheels, rims, and tires. Semitrailer shall be equipped with dual wheels on each axle. The rims and tire ratings shall conform to TRA or ZZ-T-381 recommendations for the type and size of tires furnished. Tire size, ply rating, and rim size shall be the same for all wheels on the semitrailer. Tires shall be tube or tubeless type with highway tread. When specified (see 6.2), bias ply tires shall be furnished. Tires shall be of rated capacity at least equal to the load imposed on each tire, measure at each wheel at the ground, when the semitrailer is loaded with the applicable specified payload. When specified (see 6.2), disk type wheels shall be furnished.

3.5.9.2 Inner tubes. When tube-type tires are furnished, inner tubes shall be of heavy-duty type, and shall be of proper size for the tires furnished. Tire flaps shall be provided in accordance with TRA recommendations.

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3.5.9.3 Tire carrier. One tire carrier shall be provided and installed under the frame at curb side in an accessible location. Means shall be provided for securing the tire within the carrier to prevent accidental loss.

3.5.9.4 Spare wheel. When specified (see 6.2), a spare wheel or rim shall be mounted on the tire carrier.

3.5.9.5 Spare tire. When specified (see 6.2), an inflated spare tire shall be mounted on the spare wheel or rim specified in 3.5.9.4. The spare tire shall be of the same size, tread design, and ply rating as tires installed on the semitrailer.

3.5.10 Brakes.

* 3.5.10.1 Service brakes. Service brakes shall be of the full air type and shall conform to DoT Federal Motor Carrier Safety Regulations, sections 393.45 through 393.47, and 393.52. The braking system shall include manual or automatic slack adjusters, piping, hose connections, glad hands, spring-loaded dust covers or dummy glad hands equipped with security chains, and all other components required for a complete air brake system. Glad hands shall conform to SAE J318. Air hose location shall comply with SAE J702. The braking system shall be installed in a manner which provides road clearance for travel over uneven terrain and protection against damage caused by objects striking components. No part of the braking system shall extend below the bottom of wheel rims. The brake linings shall be of the non-asbestos type.

3.5.10.2 Parking brakes. The semitrailer shall be furnished with spring or air diaphragm mechanical lock-type parking brakes. Parking brakes shall conform to DoT Federal Motor Carrier Safety Regulation, section 393.41. The parking brakes shall hold the semitrailer with rated payload on a 10 percent grade despite the depletion of the compressed air supply. The parking brakes shall be automatically applied and shall remain applied without further application.

3.5.11 Upper fifth wheel plate. The upper fifth wheel plate shall be designed for coupling to a full oscillating fore and aft rocking type fifth wheel, shall be of sufficient size to cover a fifth wheel 36 inches in diameter, and shall conform to DoT Federal Motor Carrier Safety Regulations, section 393.70(b). The kingpin shall be of heat-treated alloy steel and shall conform to SAE J700. The forward end of the upper fifth wheel plate shall have a turned up lip for ease of loading and for body protection.

* 3.5.12 Lighting. The electrical lighting system shall be 12-volt (V) potential. At least one pair of brake lights shall override the four-way emergency flasher. Modifications to the manufacturer's standard product to accommodate this requirement shall not compromise conformance to any Federal Motor Carrier Safety Regulation referenced herein or to any Federal Motor Vehicle Safety Standard. If additional lights are added to the vehicle, the lights shall be selected from the chassis manufacturer's standard matching hardware. The lighting system shall conform to DoT Federal Motor Carrier Safety Regulations, sections 393.14, 393.20, 393.22, 393.23, 393.25 through 393.29, 393.32, and 393.33. All lights and reflectors shall be protected from operational hazards by mounting in recessed or otherwise guarded locations. Lights and reflectors shall not be mounted on vertical surface of the rub rails

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or semitrailer bumpers. The front of the semitrailer shall be equipped with a receptacle conforming to SAE J560, with the receptacle conductors connected and color coded or numbered as specified herein. The receptacle shall be located in accordance with SAE J702. Turn signal lamps conforming to SAE J588, shall be furnished at the rear of the semitrailer. All electrical wiring shall conform to SAE J1292.

3.5.13 Landing gear. The semitrailer shall have two vertical lift, steel wheeled, telescopic, nonrotating, landing legs, with 2-speed gears, and a handcrank at the curb side. A holder shall be provided for the handcrank when not in use. The landing gear shall withstand, without deformation, the combined static and dynamic forces due to proportion of gross weight sustained in the forces resulting from impact during coupling and uncoupling operations. When placed in travel position, the landing gear legs shall remain positively locked. The landing gear shall be protected to preclude the entrance of foreign matter which would impair its functioning or mechanical efficiency. The landing gear shall have a range of adjustment to vary the height of the upper fifth wheel from 44 inches to not less than 50 inches from the ground. With the semitrailer coupled to a towing tractor and in level position, the clearance under the fully retracted landing gear shall exceed the semitrailer ground clearance but shall in no case be less than 11 inches.

3.5.14 Hopper bumpers. When specified (see 6.2), two heavy-duty bumpers shall be provided on each side of the body to protect hopper during loading. One bumper shall be located immediately below the hopper opening and extend the full width of hopper. The other bumper shall be located below the hopper at floor level of semitrailer and be not less than 36 inches in length.

3.5.15 Mirror. When specified (see 6.2), a mirror, not less than 7 inches in diameter, shall be mounted in such a position as to allow viewing of the contents and packer plate movement at hopper area from the engine level operator's station.

3.5.16 Rear end protection. Semitrailer rear end protection shall be in accordance with DoT Federal Motor Carrier Safety Regulations, section 393.86.

3.5.17 Rear wheel splash and stone throw protection. Rear wheels shall have rigid splash shields in front and mud flaps at rear. Splash and stone throw protection shall be in accordance with SAE J682. A metal strip not less than 1/8-inch thick and not less than one inch wide, extending the entire width of the mud flap, shall be installed to prevent the bolt heads or bolt nuts from damaging the mud flaps.

3.5.18 Toolbox. When specified (see 6.2), a sheet metal steel toolbox shall be provided to carry those tools and accessories required to operate the unit. Nominal thickness of the metal shall be 0.0598-inch (U.S. revised standard gage No. 16). The toolbox shall have a hinged lid and provisions for locking with a padlock. The toolbox shall be securely fastened to the semitrailer in a protected and accessible location.

3.6 Lubrication. Means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. Hydraulic lubrication fittings shall be in accordance

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with SAE J534. Where use of high pressure lubricating equipment, 1,000 pound-force per square inch (psi) or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location.

3.7 Treatment, painting, identification marking, and data plates. As specified by the procuring activity for the appropriate service (see 6.2), treatment, painting, identification marking, and data plates shall be in accordance with MIL-STD-1223. One coat of rust-inhibiting primer shall be applied to interior of the body. Interior of body shall be cleaned and properly prepared before applying the primer. Nameplate shall be attached to the right side of the engine housing.

* 3.8 Rustproofing. When specified (see 6.2), the semitrailer shall be rustproofed in accordance with FED-STD-297, except exterior body enclosed areas above the floor line are not required to be rustproofed.

3.9 Servicing and adjusting. Prior to acceptance of the semitrailer by the Government, the contractor shall service and adjust the semitrailer for immediate operational use as required in the operator's manual. The servicing and adjusting shall include at least the following:

a. Inflation of all tires.
b. Adjustment of brakes.
c. Proper functioning of all lighting and electrical systems.
d. Adjustment of engine to include tune-up (when required).
e. Complete lubrication with grades of lubricants recommended for ambient temperature at the delivery point.
f. Cooling system filled to capacity with a clean solution of equal parts by volume of water and antifreeze (ethylene glycol).

The semitrailer shall be conspicuously tagged to identify the lubricants and their temperature range.

3.10 Workmanship.

3.10.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.10.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.10.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or

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flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.10.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First production vehicle inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First production vehicle inspection. The first production vehicle produced under the contract shall be inspected by the contractor at his plant under the direction and in the presence of Government representatives. This inspection shall include the examination of 4.3 and the tests of 4.4. The purpose of the inspection shall be to determine vehicle conformity with the requirements of the contract. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its right under the provisions of the contract.

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4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the tests of 4.5, and the packaging MIL-S-45154I inspection of 4.6.

4.3 Examination. Each semitrailer shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 First production vehicle tests.

4.4.1 Vehicle weight test. Semitrailer shall be weighed to determine net weight and distribution of net weight on fifth wheel and axles shall be computed using the net weight and the specified payload in accordance with 3.3.4. Calculated imposed loads on fifth wheel and axles shall be utilized to ascertain that the suspension, axles, and tires furnished are of adequate capacity to meet specification requirements, and to determine conformance to 3.3.5.

4.4.2 Road test. The semitrailer loaded with the applicable payload and coupled to a truck tractor, shall be towed for a distance of not less than 100 miles, to verify the performance requirements of 3.4. At least 20 percent of this distance shall be over unimproved roads at speeds as great as 10 mph. The other 80 percent of this distance shall be over improved roads at speeds as great as 55 mph. After test, the semitrailer shall be examined for evidence of misalignment, binding or other malfunction.

4.4.3 Turning and tracking ability tests. The loaded semitrailer shall be tested for turning and tracking ability to verify compliance with the requirements of 3.4.1 and 3.4.2.

4.4.4 Landing gear test. Upon completion of the road test, function of landing gear shall be tested to verify comformance to 3.5.13.

4.4.5 Loading and compaction test. The semitrailer shall be tested for loading and compaction operation, and at least the following tests shall be made:

a. Compaction members, hydraulic system, engine, and controls shall be observed to determine conformance to 3.5.1.5, 3.5.2, and 3.5.3.
b. The body hopper capacity shall be observed to determine compliance with 3.5.1.1.
c. The hopper closing and opening operation shall be timed to determine conformance to 3.5.1.2.
d. The time required for the packer plate to be actuated through a complete cycle shall be observed to determine conformance to 3.5.1.6(a).

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4.4.6. Dumping test. The semitrailer shall be tested for compliance to 3.5.1.4 and 3.5.1.6(d).

4.4.7 Liquidtight test. Semitrailer shall be loaded with water to a depth of 5 inches to determine liquidtight capabilities specified in 3.5.1.

4.4.8 Failure. Failure of the first production vehicle to meet requirements of the contract shall be cause for the Government to refuse acceptance of all vehicles under the contract until corrective action has been taken.

4.4.9 Production sample. Upon acceptance of the first production vehicle, it shall remain at the manufacturing facility as a production sample, and may be the last semitrailer shipped on the contract. The contractor shall maintain the semitrailer in a serviceable condition for the duration of the contract.

4.5 Production semitrailer tests. The contractor's testing system shall, as a minimum, assure that the semitrailer is capable of meeting the performance requirements specified herein.

4.6 Packaging inspection. The semitrailer shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Vehicle preservation and packing. The equipment shall be preserved and packaged in accordance with the contractor's standard practice. When specified (see 6.2), equipment shall be preserved and packaged in accordance with the mobile requirements of MIL-STD-281 with the level of preservation and packaging as specified (see 6.2).

6. NOTES

6.1 Intended use. The semitrailers are intended for use in collection, transportation, and dispensing of refuse.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type of semitrailer required (see 1.2).
c. When frameless is required (see 3.5).
d. When hopper opening dimension requirements are other than as specified (see 3.5.1.1).
e. When hopper cover operation requirement is other than as specified (see 3.5.1.2)
f. When engine hour meter is not required (see 3.5.2.1).
g. When hydraulic pump discharge pressure gage is required (see 3.5.3.5).
h. When sump tank is required (see 3.5.4).

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i. When washout system is required (see 3.5.5).
j. When disk type wheels are required (see 3.5.9.1).
k. When bias ply tires are required (see 3.5.9.1).
l. When spare wheel or rim is required (see 3.5.9.4).
m. When spare tire is required (see 3.5.9.5).
n. When hopper bumpers are required (see 3.5.14).
o. When hopper area viewing mirror is required (see 3.5.15).
p. When toolbox is required (see 3.5.18).
q. Appropriate service requirements for treatment, and painting (see 3.7).
r. When rustproofing is required (see 3.8).
s. When preservation-packaging in accordance with MIL-STD-281 is required, and the level of preservation-packaging required (see 5.1).

6.3 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data shall be delivered by the contractor in accordance with the contract or purchase order requirements.

6.4 First production vehicle. The first production vehicle shall be one complete semitrailer or a standard item from the contractor's current inventory. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for examinations, tests, and approval of the first production vehicle.

6.5 Definitions.

6.5.1 Improved road. An improved road is a smooth, hard surfaced road, such as a concrete or asphalt paved highway.

6.5.2 Unimproved road. An unimproved road is an unpaved, unstabilized road with an undulating surface having occasional chuckholes and exposed rocks.

6.6 Type I and III elimination. The type I, single axle, 38 cu yd and type III, tandem axle, 42 cu yd refuse collection semitrailers were eliminated from the specification.

6.7 Subject term (key word) listing.

Engine
Hopper
Refuse

6.8 Changes from previous issue. The margins of this specifications are marked with asterisks (or vertical lines) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-45154J	MILITARY SPECIFICATION - SEMITRAILERS, REFUSE COLLECTION, COMPACTION TYPE: COMMERCIAL

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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